

Private Network Service Platform — Phase 1 Report

Computer Networks Course Project · Phase 1: Build & Observe

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Build and evidence collected on 2 October 2026 (IST) on the college Wi-Fi Rishihood_Learner.

Core principle of the project: *the application stays simple — the network is the project.*

1. Objective

Build a fully local private service on our own laptops, with no cloud, and prove every step of one request with tools:

1. A client types the private name `app.team1.test`.
2. The name is resolved by **our own DNS server** (Mac 1).
3. The client opens an **HTTPS** connection with no certificate warning to our **nginx edge** (Mac 2).
4. nginx load-balances the request to one of **two backends** (Mac 3, ports 3001 and 3002).
5. DNS, TCP, TLS and HTTP are all captured and explained with `dig`, `curl -v`, Wireshark and browser DevTools.

Phase 1 gate (spec section 5): a client resolves `app.teamX.test`, connects over HTTPS, and receives responses from both backends through the load balancer. **Status: met.** Mac 1 and Mac 3 each resolved `app.team1.test` to `10.7.16.45`, connected over HTTPS with SSL certificate verify `ok`, and received both `X-Backend: A` and `X-Backend: B`.

2. Team, roles and machine inventory (Task A)

Our team has 3 members. The spec's 4-role layout allows teams of 2–3 to combine roles (section 3), so **both backends run on Mac 3** as two separate processes on two ports.

Machine	Owner	Role	IPv4	Mask / prefix	Gateway	Interface	Wi-Fi MAC
Mac 1	Prince	Private DNS (dnsmasq) + test client	10.7.28.177	255.255.224.0 (/19)	10.7.0.1	en0	10:9f:41:bd:94:ac
Mac 2	Atanu	Edge: nginx reverse proxy, TLS termination, load balancer	10.7.16.45	255.255.224.0 (/19)	10.7.0.1	en0	hardware 80:a9:97:48:ca:2f; on this network macOS uses the private address 12:32:e6:75:62:df (seen in ifconfig, the Wi-Fi menu and Wireshark)
Mac 3	Sambhav	Backend A :3001 + Backend B	10.7.7.111	255.255.224.0 (/19)	10.7.0.1	en0	10:9f:41:c2:3a:02

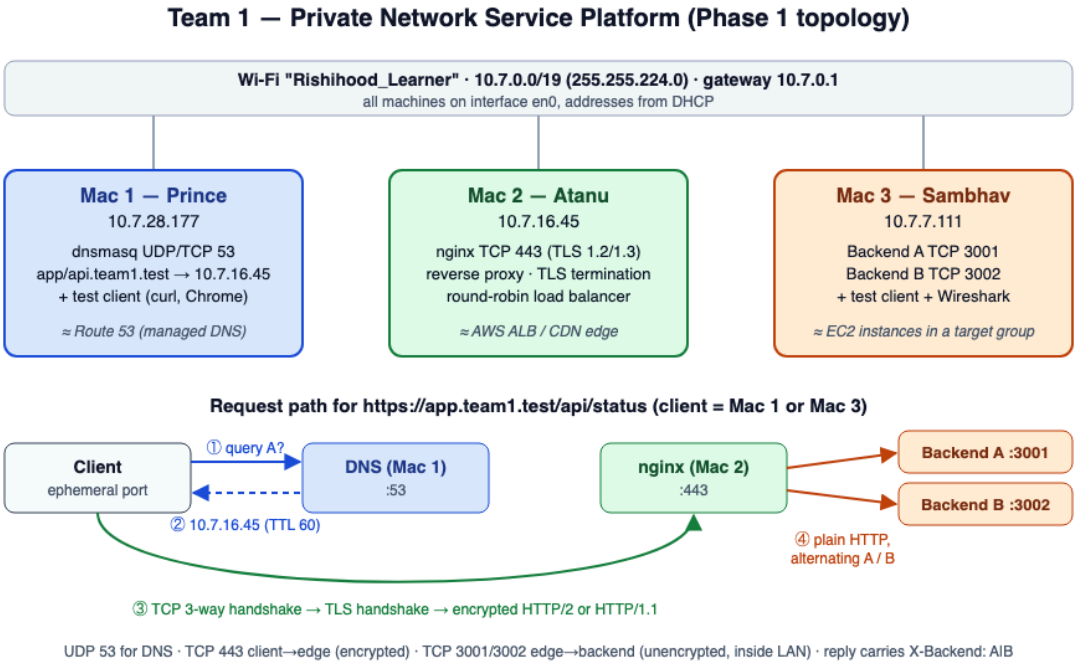
Machine	Owner	Role	IPv4	Mask / prefix	Gateway	Interface	Wi-Fi MAC
		:3002 + test client + Wireshark					

Collected with `networksetup -getinfo Wi-Fi`, `ifconfig en0` and `route -n get default` on each Mac (evidence: [evidence/A-ping](#)).

Reachability: all six directed pings (1→2, 1→3, 2→1, 2→3, 3→1, 3→2) received 4 of 4 replies, **0.0 % packet loss**, TTL 64 (same subnet, no router hop). Round-trip times ranged from about 8.6 ms to 115 ms. That spread is normal on a shared college Wi-Fi, and it also explains the TCP retransmissions in our capture (section 6).

All IPs come from DHCP. At the start of every session we run `ipconfig getifaddr en0` on all three Macs. If Mac 2’s address changes, the `dnsmasq host-record` lines must change. If Mac 3’s address changes, the `nginx upstream` must change.

3. Topology and cloud mapping



The client only ever learns the **edge's** IP (10.7.16.45). The backend addresses live only in the nginx `upstream` block. That is why we can add, remove or replace backends without changing DNS or any client.

4. Request flow, ports and protocol layers

4.1 One request, step by step (`https://app.team1.test/api/status` from Mac 3)

#	What happens	Protocol / layer	Socket (from our capture)
1	Client asks “A record for <code>app.team1.test</code> ?”	DNS over UDP	10.7.7.111: 50215 → 10.7.28.177: 53 (pkt 239)
2	dnsmasq answers 10.7.16.45, TTL 60	DNS / UDP	10.7.28.177:53 → 10.7.7.111:50215 (pkt 240, ~10 ms later)
3	ARP: “who has 10.7.16.45?” → 12:32:e6:75:62:df	ARP, link layer	broadcast (pkts 241–242)
4	TCP three-way handshake SYN → SYN-ACK → ACK	TCP	10.7.7.111: 59200 → 10.7.16.45: 443 (pkts 243–247)
5	TLS handshake: ClientHello (SNI <code>app.team1.test</code> , ALPN <code>h2/http1.1</code>) ... Finished	TLS over TCP	same connection (pkts 249–256)
6	Encrypted HTTP/2 request and response	HTTP inside TLS	Application Data (pkts 258–278)
7	nginx opens its own TCP connection to the chosen backend and forwards the request in plain HTTP/1.1	TCP + HTTP	10.7.16.45: 55239 → 10.7.7.111: 3001 (pkt 265)
8	Backend answers (HTTP/1.0, <code>X-Backend: A</code>); nginx re-encrypts toward the client	HTTP → TLS	—
9	Connection closed: Encrypted Alert, then FIN/ACK in both directions	TLS / TCP	pkts 281–286

4.2 Ports table

Service	Machine	Port / transport	Who connects
DNS (dnsmasq)	Mac 1	53 UDP (and TCP)	every client resolver
HTTPS (nginx)	Mac 2	443 TCP	clients
Backend A	Mac 3	3001 TCP	nginx only, in normal use
Backend B	Mac 3	3002 TCP	nginx only, in normal use
Client side	any	ephemeral (e.g. 50215 UDP, 59200 TCP; nginx side 55239...)	chosen by the OS per connection

A **socket pair** (src IP, src port, dst IP, dst port, protocol) identifies each connection. The client→edge connection and the edge→backend connection are **two separate TCP connections**. That is what “reverse proxy” means.

4.3 OSI vs TCP/IP mapping

OSI layer	TCP/IP layer	In our project
7 Application	Application	DNS query/answer, HTTP/1.1 and HTTP/2 (REST JSON, X-Backend, Cache-Control, ETag)
6 Presentation / 5 Session	Application (TLS sits between HTTP and TCP)	TLS 1.2 / 1.3: certificate, key exchange, encryption
4 Transport	Transport	UDP 53 for DNS; TCP 443 and TCP 3001/3002 (handshake, seq/ack, retransmission)
3 Network	Internet	IPv4 10.7.0.0/19; TTL 64 in pings
2 Data link / 1 Physical	Link	Wi-Fi 802.11ax, Ethernet II framing, MAC addresses, ARP

5. Implementation of Tasks B–F

Task B — Private DNS server (Mac 1)

dnsmasq 2.93, config at /opt/homebrew/etc/dnsmasq.conf ([copy](#), [explanation](#)):

```
interface=en0
listen-address=127.0.0.1
domain-needed
bogus-priv
local=/team1.test/
host-record=app.team1.test,10.7.16.45
host-record=api.team1.test,10.7.16.45
local-ttl=60
no-resolv
server=8.8.8.8
server=1.1.1.1
log-queries
log-facility=/tmp/dnsmasq.log
```

- `host-record` gives an exact A record for each name. Both names point to **Mac 2 (the edge)**, never to a backend.
- `local=/team1.test/` makes dnsmasq authoritative for our zone. Other names are forwarded to 8.8.8.8 / 1.1.1.1, so clients keep internet access.
- Client resolvers: Mac 2 and Mac 3 set to 10.7.28.177 (System Settings → Wi-Fi → Details → DNS, Search Domains empty). Mac 1 set to 127.0.0.1.

Results ([evidence/B-dns](#)): - Mac 2, 12:41:59 IST: `dig app.team1.test → app.team1.test. 60 IN A 10.7.16.45, SERVER: 10.7.28.177#53, flags qr aa rd ra (authoritative answer)`. - Mac 3, 12:41:59 IST: same answer from 10.7.28.177. - Mac 3, 12:46:50 IST: `dig api.team1.test → 60 IN A 10.7.16.45`. - Mac 2, 3 Oct 11:32:06 IST: `dig @8.8.8.8 app.team1.test → NXDOMAIN`, with only the root zone’s SOA in the authority

section. The name exists only inside our network. - Mac 1, 3 Oct: `sudo lsof -nP -i :53` shows dnsmasq listening on **UDP and TCP 53** on 10.7.28.177 (LAN), 127.0.0.1 (itself) and IPv6 (::1 and link-local addresses).

DNS vs connection: DNS only turns a name into an IP. It moves no application data. After the answer, the client opens a completely separate TCP connection to that IP on port 443. In the capture the DNS exchange (UDP) ends at packet 240 and the TCP SYN starts at packet 243.

Task C — Two backend services (Mac 3)

`backend/backend.py` is one Python standard-library script started twice: `python3 backend.py A 3001` and `python3 backend.py B 3002`.

Endpoint	Response
GET /	<code>{"service":"team1 backend","backend":"A","message":"Backend A is running"}</code>
GET /api/status	<code>{"backend":"A","status":"ok","time":"..."}</code> , Cache-Control: no-store
GET /api/info	Same JSON on both backends, Cache-Control: public, max-age=60, ETag: "6c623c0954f8001c", 304 on a matching If-None-Match
every response	X-Backend: A or X-Backend: B

Both bind to 0.0.0.0, so `lsof` on Mac 3 shows `TCP *:3001 (LISTEN)` and `TCP *:3002 (LISTEN)` (not 127.0.0.1 only). `/usr/bin/curl -i http://localhost:3001/api/status` returns `HTTP/1.0 200 OK, Server: TeamBackend/1.0 Python/3.12.4, X-Backend: A`. Mac 2 reached both over the LAN: `curl -i http://10.7.7.111:3001/api/status` → `X-Backend: A`, and `:3002` → `X-Backend: B`. These checks were re-run on 3 Oct 2026, with the outputs in the evidence folder. The backend terminal logs show requests from 127.0.0.1 (local test) and from 10.7.16.45 (nginx) ([evidence/C-backends](#)).

Task D — Edge reverse proxy and load balancer (Mac 2)

nginx 1.31.6. The server block is in `/opt/homebrew/etc/nginx/servers/team1.conf` ([copy](#)):

```
log_format team1 '$remote_addr -> $upstream_addr [$time_local] "$request" $status';

upstream team1_backends {
    server 10.7.7.111:3001 max_fails=1 fail_timeout=10s;    # Backend A
    server 10.7.7.111:3002 max_fails=1 fail_timeout=10s;    # Backend B
}

server {
    listen 443 ssl;
    http2 on;
    server_name app.team1.test api.team1.test;
    ssl_certificate      /opt/homebrew/etc/nginx/certs/app.team1.test.crt;
    ssl_certificate_key  /opt/homebrew/etc/nginx/certs/app.team1.test.key;
    ssl_protocols       TLSv1.2 TLSv1.3;
    access_log /opt/homebrew/var/log/nginx/team1_access.log team1;

    location / {
        proxy_pass http://team1_backends;
        proxy_set_header Host $host;
```

```

    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto https;
    proxy_connect_timeout 2s;
    proxy_next_upstream error timeout http_502 http_503;
}
}

```

- **Strategy:** round-robin (nginx's default). Each new request goes to the next server in the list.
- **Passive health check:** after 1 failed attempt a server is marked down for 10 s. `proxy_next_upstream` retries the failed request on the other server, so the client still gets a 200.
- **Results:** a loop of 6 requests from Mac 1, Mac 2 and Mac 3 printed A, B, A, B, ... The nginx access log ([evidence/D-loadbalancing](#)) holds the requests from all three clients made during the build on 2 Oct: **62 served by :3001 and 61 by :3002**, for example:

```

10.7.7.111 -> 10.7.7.111:3002 [02/Oct/2026:13:44:54 +0530] "GET /api/status HTTP/2.0" 200
10.7.7.111 -> 10.7.7.111:3001 [02/Oct/2026:13:44:54 +0530] "GET /api/status HTTP/2.0" 200
10.7.7.111 -> 10.7.7.111:3002 [02/Oct/2026:13:44:55 +0530] "GET /api/status HTTP/2.0" 200

```

Why the client never needs backend IPs: DNS gives out only the edge address. The client's TCP connection ends at nginx, and nginx alone decides which backend to use. The client can see which backend answered only because of the `X-Backend` header we added on purpose.

Task E — HTTPS / TLS

We used our **own team CA** instead of a bare self-signed certificate, so every client can trust it properly without `-k` ([setup notes](#)).

Item	Value
CA	CN=Team1 Local CA, O=Team1 CN Project, C=IN; RSA 2048, CA:TRUE, keyCertSign
Server certificate	CN=app.team1.test, issued by Team1 Local CA
SAN	DNS:app.team1.test, DNS:api.team1.test
EKU / validity	serverAuth; 2 Oct 2026 07:40:15 GMT → 2 Oct 2027
Check	<code>openssl verify -CAfile team1-ca.crt app.team1.test.crt</code> → OK
Trust	<code>team1-ca.crt</code> added as <code>trustRoot</code> to the System keychain on all three Macs ; private keys never left Mac 2
nginx	TLS terminates at nginx on 443, TLS 1.2 + 1.3, HTTP/2 enabled

Results ([evidence/E-tls](#)), from `/usr/bin/curl -v` on Mac 3:

	TLS 1.2 (<code>--tls-max 1.2</code>)	TLS 1.3 (default)
Handshake seen by curl	Client hello → Server hello → Certificate → Server key exchange → Server finished → Client key exchange → Change cipher spec → Finished	Client hello → Server hello → Encrypted Extensions ("Unknown (8)") → Certificate → CERT verify → Finished

	TLS 1.2 (<code>--tls-max 1.2</code>)	TLS 1.3 (default)
Negotiated	TLSv1.2 / ECDHE-RSA-CHACHA20-POLY1305	TLSv1.3 / AEAD-CHACHA20-POLY1305-SHA256
ALPN	server accepted h2	server accepted h2
Verification	subjectAltName: host "app.team1.test" matched · SSL certificate verify ok	same
Response	HTTP/2 200, x-backend: A	HTTP/2 200, x-backend: B

Also shown: Mac 1 `curl -v https://app.team1.test` (TLS 1.3, verify ok, HTTP/2 200), the browser opening `https://app.team1.test/api/status` by name with no warning, `curl -sI --http1.1` → HTTP/1.1 200 OK (with Connection: keep-alive) and `--http2` → HTTP/2 200 (lower-case header names, as HTTP/2 requires) on 3 Oct. The browser's **Certificate Viewer** (Chromium-based Brave on a client Mac) shows *Issued To* `app.team1.test / Team1 CN Project`, *Issued By* `Team1 Local CA`, valid Friday 2 October 2026 13:10:15 → Saturday 2 October 2027 13:10:15 IST (= 07:40:15 GMT), and SHA-256 fingerprint `73533e1c...b358`. **HTTP/3** (QUIC over UDP) is explanation-only, as the spec allows.

Handshake in words: ClientHello (supported versions and ciphers, SNI, ALPN, key share) → ServerHello (chosen cipher; in our capture `0xcc8` for TLS 1.2 and `0x1303` for TLS 1.3) → Certificate (server proves its identity with a chain to a CA the client trusts) → Key Exchange (ECDHE, so both sides derive the same session keys) → ChangeCipherSpec / Finished (everything after this is encrypted). In TLS 1.3 the certificate itself is already encrypted, which is why we also captured a TLS 1.2 session to see the Certificate in clear.

Task F — HTTP caching

`/api/info` returns `Cache-Control: public, max-age=60` and an ETag built from a hash of the body, so Backend A and Backend B produce the **same** ETag ([evidence/F-caching](#)).

```
$ curl -I https://app.team1.test/api/info (Mac 1, 11:41:03 GMT)
HTTP/2 200
server: nginx/1.31.6
content-type: application/json
content-length: 117
x-backend: B
cache-control: public, max-age=60
etag: "6c623c0954f8001c"

$ curl -i -H "If-None-Match: $ETAG" https://app.team1.test/api/info
HTTP/2 304 (x-backend: A at 11:42:31 and 11:45:45, x-backend: B at 11:45:48)
```

- The **304 Not Modified** came from both A and B. Round-robin does not break revalidation because the ETag is content-based.
- **Chrome DevTools:** a repeated `fetch` was served 200 (disk cache) in **1 ms** (no network at all). Reloading the page sent a conditional request that got **304** (0.1 kB).
- **nginx log** shows the same pattern: `"GET /api/info HTTP/2.0" 304` lines, plus 200 when the cache was empty or expired.
- `/api/status` uses `Cache-Control: no-store`, so live status is never cached.

Case	What happens on the network	Our evidence
Fresh cache hit	Response younger than max-age → served locally, no request sent	DevTools (disk cache), 1 ms
Conditional request	Copy expired or reload → client sends If-None-Match, server replies 304 with headers only	curl 304, DevTools 304
Full new request	No cached copy → full 200 with body	curl -I 200, log 200

6. Task G — Packet capture analysis

Captured on **Mac 3, Wi-Fi en0**, no capture filter. Saved as [phase1_full_flow.pcapng](#) (**683 packets, 483 s**). Because Mac 3 is both the client and the backend host, the same file also holds all six nginx→backend connections, in plain text. A machine-readable summary made with `tshark` is in [phase1_full_flow_tshark_summary.txt](#). Traffic: DNS cache flushed, then `/usr/bin/curl -v --tls-max 1.2 .../api/status`, then TLS 1.3, then a 4-request loop ([evidence/G-capture](#)).

Layer / event	Display filter	What we found
DNS	<code>dns.qry.name contains "team1.test"</code>	Pkt 239 (t = 101.203 s) standard query 0x647b A app.team1.test, UDP 50215 → 53 . Pkt 240 (t = 101.214 s) response A 10.7.16.45, TTL 60 (1 minute) , flags 0x8580 (response, authoritative, no error), answered 10.27 ms after the request (screenshot). Second lookup pkts 392/393 (0x5868). Conversations: all 210 UDP conversations of Mac 3 go to 10.7.28.177:53. The team1.test ones use source ports 50215 and 51388
ARP	—	Pkt 241 “Who has 10.7.16.45? Tell 10.7.7.111”, pkt 242 “10.7.16.45 is at 12:32:e6:75:62:df”: IP → MAC before the first frame to the edge
TCP handshake	<code>tcp.flags.syn==1, tcp.stream eq 0</code>	Pkt 243 SYN 59200 → 443, Seq=0, Win=65535, MSS=1460, WS=64, SACK_PERM · pkt 244 SYN retransmission · pkt 245 SYN-ACK Seq=0 Ack=1 · pkt 246 duplicate SYN-ACK · pkt 247 ACK Seq=1 Ack=1 · pkt 248 Dup ACK. Only then does TLS start (pkt 249)
TLS 1.2	<code>tcp.stream eq 0 && tls</code>	249 Client Hello (SNI=app.team1.test, 46 cipher suites, ALPN h2/http/1.1) · 252 Server Hello, Certificate, Server Key Exchange, Server Hello Done (tls.handshake.type == 11 → only pkt 252) · 254 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message · 256 Change Cipher Spec, Encrypted Handshake

Layer / event	Display filter	What we found
		Message · 258–278 Application Data · 281 Encrypted Alert
TLS 1.3	<code>tcp.stream eq 2 && tls</code>	Pkt 399 Client Hello from port 59239 (TLS 1.3 suites incl. TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256) · 402 Server Hello, Change Cipher Spec, Application Data. The certificate is no longer visible because TLS 1.3 encrypts it
Encrypted payload	<code>tls.record.content_type == 23</code>	Pkts 258, 259, 260, 262, 266, 278 “Application Data”. The hex pane shows unreadable bytes, so the HTTP headers that <code>curl -v</code> prints are not visible on the wire
Edge → backend	<code>tcp.port == 3001 or tcp.port == 3002, Follow TCP Stream</code>	Pkt 265 nginx SYN 10.7.16.45:55239 → 10.7.7.111:3001; request pkt 269, response pkt 274. Follow TCP Stream shows plain text : GET /api/status HTTP/1.1, Host: app.team1.test, X-Real-IP: 10.7.7.111, X-Forwarded-Proto: https → HTTP/1.0 200 OK, Server: TeamBackend/1.0 Python/3.12.4, X-Backend: A. The six backend responses (pkts 274, 425, 505, 545, 585, 626) carry X-Backend A, B, A, B, A, B . This proves TLS terminates at the edge
Load balancing	Statistics → Conversations → TCP (12)	Six client→443 connections (source ports 59200, 59239, 59247–59250) and six nginx→backend connections (55239, 55245–55249) alternating 3001 / 3002
Teardown	<code>tcp.stream eq 0</code>	Pkts 282–286 FIN/ACK in both directions (client Seq=524, Ack=1713)

Sequence / acknowledgement numbers (reliability): the numbers are relative. SYN uses Seq 0. Each side’s ACK number is “the next byte I expect”. After the ClientHello (231 bytes of TCP payload, Seq=1), the server acknowledges **Ack=232**. After the server’s 1381-byte flight the client acknowledges Ack=1382. By the end the client has sent 523 bytes (Seq=524) and received 1712 (Ack=1713). Pkts 244 and 246 are **retransmissions**: on this busy Wi-Fi the first SYN/SYN-ACK was not acknowledged in time, so TCP re-sent it. The connection still completed, which is TCP’s reliable delivery at work and not an error. The `Win=` field (e.g. 131712 after window scaling) is TCP **flow control**: how many bytes the receiver can accept.

7. Phase 1 failure demonstrations (spec 6.3)

The submission form (D3) asks for **one** failure scenario demonstrated fully. We chose **Option A: stop one backend**, which is also demo step 8 in spec section 8. It was recorded on 3 Oct 2026 and is described in 7.1.

Section 7.2 lists all five scenarios from spec 6.3, with the procedure for the ones we have prepared but not yet recorded, in case the evaluator asks for them live.

7.1 Recorded demonstration — Option A: stop Backend A (3 Oct 2026, IST)

Evidence: [evidence/failures/3-one-backend-stopped](#). It holds the terminal outputs from all three Macs and the matching nginx log lines exported from Mac 2.

Before (11:12:40). Mac 1 ran six requests:

```
$ for i in 1 2 3 4 5 6; do curl -s -i https://app.team1.test/api/status | grep -i x-backend; done
x-backend: A    x-backend: A    x-backend: B    x-backend: A    x-backend: B    x-backend: A
```

The Mac 2 access log shows the same order (3001, 3001, 3002, 3001, 3002, 3001). Both backends are serving. The first two requests both went to A because nginx's round-robin position carries over from earlier traffic, so a new loop does not always start on a clean A/B boundary. After that the order alternates.

Break. On Mac 3 we pressed Ctrl+C in the Backend A window. `lsof -nP -iTCP -sTCP:LISTEN | grep -E ":3001|:3002"` then listed only `*:3002 (LISTEN)`.

After (11:14:40–41). The same loop on Mac 1 returned:

```
x-backend: B    x-backend: B    x-backend: B    x-backend: B    x-backend: B    x-backend: B
```

Every request still got HTTP 200, so the user saw no error. nginx's two logs on Mac 2 explain why:

```
error.log  2026/10/03 11:14:41 [error] ... connect() failed (61: Connection refused) while
connecting to upstream,
            client: 10.7.28.177, ... upstream: "http://10.7.7.111:3001/api/status"
access.log 10.7.28.177 -> 10.7.7.111:3001, 10.7.7.111:3002 [03/Oct/2026:11:14:41 +0530] "GET
/api/status HTTP/2.0" 200
access.log 10.7.28.177 -> 10.7.7.111:3002 [03/Oct/2026:11:14:41 +0530] "GET /api/status HTTP/2.0"
200    (x4 more)
```

The access-log line with **two** upstreams shows one request first tried port 3001, was refused, and was retried on 3002 (`proxy_next_upstream error`). Because of `max_fails=1 fail_timeout=10s`, nginx then marked A as down and sent the rest directly to B.

To prove the machine itself was still up, we ran two commands on Mac 2:

```
$ curl -i http://10.7.7.111:3001/api/status
curl: (7) Failed to connect to 10.7.7.111 port 3001 after 1072 ms: Couldn't connect to server
$ ping -c 2 10.7.7.111
2 packets transmitted, 2 packets received, 0.0% packet loss
```

Layer affected. The **application layer on the backend side**: the Backend A process stopped, so nothing listened on TCP port 3001. Every layer below it still worked:

- DNS still resolved `app.team1.test` to 10.7.16.45.
- The client's TCP + TLS connection to nginx on port 443 still succeeded.
- Mac 3 still answered ping (the IP layer was fine).

Only nginx's own TCP connection to port 3001 was refused: Mac 3's operating system sent a TCP RST, error 61. The client noticed nothing, because it only ever talks to the edge. The edge's health check hid the failure.

Restore (11:17:51). On Mac 3, `python3 backend.py A 3001` printed `Backend A listening on 0.0.0.0:3001`. Its log immediately showed requests from nginx (`10.7.16.45 ... "GET /api/status HTTP/1.1" 200`). The loop on Mac 1 returned `B, B, A, B, A, B`, and the access log matches (`3002, 3002, 3001, 3002, 3001, 3002`). Backend A is back in the rotation.

7.2 All five scenarios from spec 6.3

#	Scenario	How we break it	Expected observation	Restore	Status
1	Wrong DNS server on a client	Mac 3: <code>sudo networksetup -setdnsservers Wi-Fi 8.8.8.8 + flush</code>	<code>dig app.team1.test</code> → NXDOMAIN from 8.8.8.8. <code>curl</code> → "Could not resolve host". But ping 10.7.16.45 works. DNS and IP connectivity are independent layers	set back to <code>10.7.28.177</code> + flush	Seen by accident during setup (Mac 2/3 had no DNS set → NXDOMAIN via 8.8.8.8); deliberate run prepared, not recorded
2	DNS record → wrong IP	Mac 1: change <code>host-record=app.team1.test,...</code> to <code>10.7.7.111</code> , restart <code>dnsmasq</code> , flush client	<code>dig</code> succeeds (NOERROR, answer 10.7.7.111), but <code>curl https://app.team1.test</code> → "Failed to connect ... port 443: Connection refused", because nothing listens on 443 there. DNS is a directory, not a connection	put back <code>10.7.16.45</code> , restart, flush	Wrong answer <code>10.7.7.111</code> captured by accident at 12:12 IST on Mac 2 and Mac 3 (evidence); <code>curl</code> consequence prepared, not recorded
3	One backend stopped (our form choice: Option A)	Mac 3: <code>Ctrl+C</code> Backend A	Loop returned 200, only X-Backend: B . <code>error.log</code> <code>connect() failed (61: Connection refused) ... 10.7.7.111:3001</code> , and access log <code>3001, 3002 ... 200</code> shows the retry. From Mac 2, <code>curl http://10.7.7.111:3001</code> failed while <code>ping 10.7.7.111</code> worked	<code>python3 backend.py A 3001</code> → <code>B, B, A, B, A, B</code>	✅ Recorded 3 Oct 2026, see 7.1
4	Both backends stopped	Mac 3: <code>Ctrl+C</code> A and B	<code>curl -v</code> : DNS ✓, TCP ✓, TLS ✓ (verify ok), then HTTP/2 502 generated by nginx itself: no X-Backend header, <code>server: nginx</code> . This is where the edge ends and the backend begins	restart both	Prepared, not recorded
5	Wrong destination	Client: <code>curl -v https://app.team1.test:9443</code>	"Failed to connect to <code>app.team1.test</code> port 9443 ...	—	Prepared, not recorded

#	Scenario	How we break it	Expected observation	Restore	Status
	port		Connection refused" (a TCP RST). <code>nc -vz app.team1.test 443</code> succeeds, 9443 is refused, ping works. IP finds the host, the port finds the service		

Related unplanned observation (already in our Mac 2 logs, 2 Oct 23:52 IST): when Mac 3 was not reachable at all, nginx logged `upstream timed out (60: Operation timed out)` while connecting to upstream for **both** 3002 and 3001, then returned **504 Gateway Timeout**. Compare with scenario 4: a backend host that is *up but has no process on the port* answers with a TCP RST, so nginx gives **502 Bad Gateway** at once. A host that is *gone* sends nothing, so nginx waits for `proxy_connect_timeout` (2 s per server) and gives **504**.

8. Problems we hit and how we fixed them

#	Problem	Root cause	Fix / lesson
1	Could not clear the ~700-line default <code>dnsmasq.conf</code> in nano	Wrong tool for a full rewrite	Overwrote it with <code>cat > file <<'EOF'</code>
2	<code>dig</code> still returned the old IP after an edit	<code>brew services start</code> does nothing when the service is already running	Always <code>sudo brew services restart dnsmasq</code>
3	DNS answered 10.7.7.111 (Mac 3) instead of the edge	Wrong IP typed into <code>host-record</code> , and it was accepted without cross-checking	Confirmed Mac 2 = 10.7.16.45 with <code>ipconfig getifaddr en0</code> , fixed, re-verified from every Mac. Kept as failure evidence (scenario 2)
4	Mac 2/3 <code>dig</code> → NXDOMAIN with <code>SERVER: 8.8.8.8</code>	Their system DNS had never been set	<code>networksetup -setdnsservers Wi-Fi 10.7.28.177</code> + cache flush
5	8.8.4.4 appeared under Search Domains on one Mac	IP typed in the wrong box	Removed it. Search Domains stay empty
6	Error reading file <code>~/Downloads/team1-ca.crt</code> on Mac 2	Ran the Mac 1/3 command; on Mac 2 the CA is in <code>~/cn-project/certs</code>	Used the local path
7	Mac 1 <code>curl</code> → "Could not resolve host" while <code>dig @127.0.0.1</code> worked	<code>dig @server</code> bypasses the system resolver, and Mac 1's own system DNS was empty	Set Mac 1 DNS to 127.0.0.1. Lesson: test with plain <code>dig / dscacheutil</code> , which is what curl and browsers use
8	Mac 3 load-balancing loop printed nothing	<code>curl</code> was Anaconda's (<code>/opt/anaconda3/bin/curl</code>), which ignores the macOS Keychain; <code>-s</code> hid the certificate error	Use <code>/usr/bin/curl</code> on Mac 3

#	Problem	Root cause	Fix / lesson
9	Chrome on Mac 1: ERR_ADDRESS_UNREACHABLE	macOS Local Network privacy permission for Chrome	Privacy & Security → Local Network → Chrome ON, then restart Chrome
10	Wireshark filter bar turned red	Capture-filter syntax (host ...) typed as a display filter	Display filter ip.addr == ...
11	Conversations showed 1 row and no ports	“Limit to display filter” ticked; window scrolled right	Unticked, scrolled left
12	dns filter showed only Google/Apple lookups	Mac 3 sends all its DNS to Mac 1	dns.qry.name contains "team1.test"
13	Red retransmission rows in the handshake	Real Wi-Fi delay	Not a bug. Used as TCP reliability evidence

9. Key learnings

- **Layers really are independent.** DNS working says nothing about TCP. TCP working says nothing about TLS. TLS working says nothing about the backend. Our debugging always went name resolution → TCP connect → TLS → HTTP.
- **DNS is a directory, not a connection.** A wrong record still “works” at the DNS layer. It just sends you to the wrong place.
- **The edge hides the backends.** One name, one IP and one certificate for the client. Any number of backends, health checks and plain HTTP stay behind it.
- **TLS termination** means encryption ends at nginx. Wireshark proved it: ciphertext on 443, readable HTTP on 3001/3002. Trust comes from the CA being in each client’s keychain, not from the certificate alone.
- **Caching** saves round trips (fresh hit, no request at all) or bandwidth (304, headers only). A content-based ETag keeps revalidation working behind a load balancer.
- **Remaining single point of failure:** nginx on Mac 2, and also dnsmasq on Mac 1. Removing them is Phase 2 (backup DNS, standby edge with DNS cutover).

10. Phase 1 checklist

Task	Status
A – LAN (IPs, masks, gateway, interface, MAC, pings, topology diagram)	✓
B – Private DNS, 2 client Macs resolving through Mac 1, app + api records	✓
C – Two backends on LAN interface, /, /api/status, X-Backend	✓
D – nginx round-robin, A/B alternation proven	✓
E – HTTPS with team CA, trusted on all clients, no -k, HTTP/1.1 + HTTP/2	✓
F – Cache-Control, ETag, 304, disk-cache hit	✓
G – DNS / TCP / TLS / encrypted data / ports / LB captured	✓

Task	Status
D3 / demo step 8 – Option A failure demo (one backend stopped)	✅ recorded 3 Oct 2026
6.3 – Other four failure scenarios	procedure prepared (section 7.2); show live if the evaluator asks
Faculty confirmation of team number (<code>team1</code>) and the 3-machine role split	■

Appendix — repository map

Deliverable (spec section 9)	Where
Architecture document	this report + diagrams/topology.svg
Configuration bundle	config/ : dnsmasq, nginx, TLS notes, client DNS setup
Backend source code + launch instructions	backend/ , helper scripts in scripts/
Evidence folder	evidence/ , organised A–G + failures